EC200U&EG91xU Series QuecOpen(SDK) Low Power Consumption API Reference Manual

LTE Standard Module Series

Version: 1.1

Date: 2023-10-10

Status: Released

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About the Document

Revision History

Version	Date	Author	Description
-	2021-10-08	Kevin WANG	Creation of the document
1.0	2021-10-21	Kevin WANG	First official release
1.1	2023-10-10	Felix Lin	1. Updated the document name based on the unified QuecOpen naming. 2. Added the applicable modules EG915U series and EG912U-GL. 3. Added the APIs to set and get the time for UART to enter into sleep after no data interaction (Chapters 2.3.9 and 2.3.10).

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1 Introduction

Quectel EC200U series and EG91xU family modules support QuecOpen® solution. QuecOpen is an embedded development platform based on RTOS, which is intended to simplify the design and development of IoT applications. For more information on QuecOpen®, see ***document [1]***.

This document is applicable to QuecOpen® solution based on CSDK build environment. It mainly introduces the API functions and debugging methods related to low power consumption function of EC200U series and EG91xU family modules.

1.1. Applicable Modules

Table 1: Applicable Modules

Module Family	Module
-	EC200U Series
EG91xU	EG912U-GL
	EG915U Series

2 Low Power Consumption API

2.1. Header File

ql_power.h is the header file of low power consumption API, and *ql_uart.h* is the header file of UART sleep API without data interaction. They are all located in the directory of *ql-sdk\ql-components\ql-kernel\inc.* Unless otherwise specified, all header files mentioned in this document are in this directory.

2.2. API Overview

Table 2: API Overview

Function	Description
ql_autosleep_enable()	Enables/disables the system to sleep.
ql_autosleepex_enable()	Enables enhanced sleep.
ql_lpm_wakelock_create()	Creates a wake lock.
ql_lpm_wakelock_delete()	Deletes the created wake lock.
ql_lpm_wakelock_lock()	Locks the wake lock.
ql_lpm_wakelock_unlock()	Unlocks the wake lock.
ql_sleep_register_cb()	Registers a sleep callback function.
ql_wakeup_register_cb()	Registers a wake callback function.
ql_uart_set_sleep_delay_time()	Sets the time for UART to enter into sleep after no data interaction.
ql_uart_get_sleep_delay_time()	Gets the time for UART to enter into sleep after no data interaction.

2.3. API Description

2.3.1. ql_autosleep_enable

This function enables/disables the system to sleep.

- **Prototype**

```
ql_errcode_sleep ql_autosleep_enable(QL_SLEEP_FLAG_E sleep_flag)
```

- **Parameter**

sleep_flag:
[In] The flag of enabling or disabling the system to sleep. See ***Chapter 2.3.1.2*** for details.

- **Return Value**

See ***Chapter 2.3.1.1*** for error codes.

2.3.1.1. ql_errcode_sleep

```
typedef enum
{
QL_SLEEP_SUCCESS                = QL_SUCCESS,
QL_SLEEP_INVALID_PARAM          = (QL_COMPONENT_PM_SLEEP << 16) | 1000,
QL_SLEEP_LOCK_CREATE_FAIL       = (QL_COMPONENT_PM_SLEEP << 16) | 1001,
QL_SLEEP_LOCK_DELETE_FAIL       = (QL_COMPONENT_PM_SLEEP << 16) | 1002,
QL_SLEEP_LOCK_LOCK_FAIL         = (QL_COMPONENT_PM_SLEEP << 16) | 1003,
QL_SLEEP_LOCK_UNLOCK_FAIL       = (QL_COMPONENT_PM_SLEEP << 16) | 1004,
QL_SLEEP_LOCK_AUTOSLEEP_FAIL = (QL_COMPONENT_PM_SLEEP << 16) | 1005,
QL_SLEEP_PARAM_SAVE_FAIL        = (QL_COMPONENT_PM_SLEEP << 16) | 1006,
}ql_errcode_sleep
```

- **Member**

Member	Description
QL_SLEEP_SUCCESS	Successful execution
QL_SLEEP_INVALID_PARAM	Invalid parameter
QL_SLEEP_LOCK_CREATE_FAI	Failed to create a sleep lock

QL_SLEEP_LOCK_DELETE_FAIL	Failed to delete a sleep lock
QL_SLEEP_LOCK_LOCK_FAIL	Failed to lock a sleep lock
QL_SLEEP_LOCK_UNLOCK_FAIL	Failed to unlock a sleep lock
QL_SLEEP_LOCK_AUTOSLEEP_FAIL	Failed to start automatic sleep
QL_SLEEP_PARAM_SAVE_FAIL	Failed to save the parameter

2.3.1.2. QL_SLEEP_FLAG_E

```
typedef enum
{
  QL_NOT_ALLOW_SLEEP = 0,
  QL_ALLOW_SLEEP,
}QL_SLEEP_FLAG_E
```

- **Member**

Member	Description
QL_NOT_ALLOW_SLEEP	Disable the system to sleep
QL_ALLOW_SLEEP	Enable the system to sleep

2.3.2. ql_autosleepex_enable

This function enables enhanced sleep.

- **Prototype**

```
ql_errcode_sleep ql_autosleepex_enable(QL_SLEEP_FLAG_E sleep_flag, uint8_t no_data_time, uint16_t punish_time)
```

- **Parameter**

sleep_flag:
[In] The flag of enabling or disabling the system to sleep. See ***Chapter 2.3.1.2*** for details.

no_data_time:
[In] The duration of releasing RRC after no data interaction. Unit: s.

punish_time:
[In] The duration of restoring the enhanced sleep mode if an abnormal network registration is encountered.
Unit: s.

- **Return Value**

See ***Chapter 2.3.1.1*** for error codes.

2.3.3. ql_lpm_wakelock_create

This function creates a wake lock.

- **Prototype**

```
int ql_lpm_wakelock_create(char *lock_name, int name_size)
```

- **Parameter**

lock_name:
[In] The name of a wake lock. The maximum length is 32 bytes.

name_size:
[In] Invalid parameter.

- **Return Value**

Less than or equal to 0	Failed execution
More than 0	Successful execution. Returns the handle of the lock, which is passed in by subsequent operations.

2.3.4. ql_lpm_wakelock_delete

This function deletes the created wake lock.

- **Prototype**

```
ql_errcode_sleep ql_lpm_wakelock_delete(int wakelock_fd)
```

- **Parameter**

wakelock_fd:
[In] Handle of the wake lock. Returned by *ql_lpm_wakelock_create()*.

- **Return Value**

See ***Chapter 2.3.1.1*** for error codes.

2.3.5. ql_lpm_wakelock_lock

This function locks the wake lock. If there are one or more wake locks in the system, the system is not allowed to sleep.

- **Prototype**

```
ql_errcode_sleep ql_lpm_wakelock_lock(int wakelock_fd)
```

- **Parameter**

wakelock_fd:
[In] Handle of the wake lock. Returned by *ql_lpm_wakelock_create()*.

- **Return Value**

See ***Chapter 2.3.1.1*** for error codes.

2.3.6. ql_lpm_wakelock_unlock

This function unlocks the wake lock. Only when all wake locks in the system are unlocked can the system sleep.

- **Prototype**

```
ql_errcode_sleep ql_lpm_wakelock_unlock(int wakelock_fd)
```

- **Parameter**

wakelock_fd:
[In] Handle of the wake lock. Returned by *ql_lpm_wakelock_create()*.

- **Return Value**

See ***Chapter 2.3.1.1*** for error codes.

2.3.7. ql_sleep_register_cb

This function registers a sleep callback function.

- **Prototype**

```
ql_errcode_sleep ql_sleep_register_cb(ql_enter_sleep_callback cb)
```

- **Parameter**

cb:
[In] Sleep callback function pointer.

- **Return Value**

See ***Chapter 2.3.1.1*** for error codes.

2.3.8. ql_wakeup_register_cb

This function registers a wakeup callback function.

- **Prototype**

```
ql_errcode_sleep ql_wakeup_register_cb(ql_exit_sleep_callback cb)
```

- **Parameter**

cb:
[In] Wakeup callback function pointer.

- **Return Value**

See ***Chapter 2.3.1.1*** for error codes.

2.3.9. ql_uart_set_sleep_delay_time

This function sets the time for UART to enter into sleep after no data interaction.

- **Prototype**

```
ql_uart_errcode_e ql_uart_set_sleep_delay_time(uint8_t times)
```

- **Parameter**

times:
[In] The time for UART to enter into sleep after no data interaction. Unit: s. The minimum value: 1; the maximum value: 255.

- **Return Value**

See ***Chapter 2.3.9.1*** for error codes.

2.3.9.1. ql_uart_errcode_e

The return value of UART sleep API without data interaction indicates whether the function is executed successfully. If not, the error reason is returned. The enumeration of UART error codes is defined as below:

```
typedef enum
{
    QL_UART_SUCCESS                 = 0,
    QL_UART_EXECUTE_ERR             = 1|QL_UART_ERRCODE_BASE,
    QL_UART_MEM_ADDR_NULL_ERR,
    QL_UART_INVALID_PARAM_ERR,
    QL_UART_OPEN_REPEAT_ERR,
    QL_UART_NOT_OPEN_ERR            = 5|QL_UART_ERRCODE_BASE,
} ql_uart_errcode_e
```

- **Member**

Member	Description
QL_UART_SUCCESS	Successful execution
QL_UART_EXECUTE_ERR	Failed execution
QL_UART_MEM_ADDR_NULL_ERR	The parameter address is NULL
QL_UART_INVALID_PARAM_ERR	Invalid parameter

QL_UART_OPEN_REPEAT_ERR	UART repeatedly opened
QL_UART_NOT_OPEN_ERR	UART is not open

2.3.10. ql_uart_get_sleep_delay_time

This function gets the time for UART to enter into sleep after no data interaction.

- **Prototype**

```
ql_uart_errcode_e ql_uart_get_sleep_delay_time(uint8_t *times)
```

- **Parameter**

times:
[Out] The time for UART to enter into sleep after no data interaction. Unit: s.

- **Return Value**

See ***Chapter 2.3.9.1*** for error codes.

3 Demo and Debugging

This chapter mainly introduces how to use the above API functions in Kernel and Application, and debug the low power consumption function.

3.1. Demo Introduction

power_demo.c, the Application demo, is located in the directory of *\components\ql-application\power* of CSDK. Two sleep locks and a timer are created with power task in the demo code. After the timer runs for 1 second, these two wake locks will be released and the system is enabled to sleep.

3.2. Low Power Consumption Debugging

3.2.1. Sleep Failure Analysis

1. Only when all conditions below are met can the system sleep:

   - USB device is plugged out.
   - All system wake locks are unlocked.
   - No external pin interruption or interference.
   - The system is enabled to automatic sleep.
   - **AT+CFUN=0** is set or the system is in a normal network registration state.

2. Capture the log for failing to enter into sleep mode.

If all the conditions mentioned above are met but the system still cannot sleep for a long time, the log indicating the reason why the system cannot sleep will be printed out every 3 seconds, as shown in the figure below. Please save the log and submit it to Quectel Technical Supports for further analysis.

13:59:18.823	15879	QOPN/I	[ql_POWER][ql_power_demo_thread, 88] receive event, id is -2015266383
13:59:18.823	15880	QUEC/I	[quec_power_mg][quec_pmsource_auto_sleep, 274] start auto sleep mode
13:59:18.823	15882	QOPN/I	[ql_POWER][ql_lpm_wakelock_unlock, 337] unlock ok
13:59:18.823	15882	QOPN/I	[ql_POWER][ql_lpm_wakelock_unlock, 337] unlock ok
13:59:18.823	15882	QOPN/I	[ql_POWER][ql_power_demo_thread, 83] set auto sleep mode ok
13:59:21.810	65035	QUEC/I	[quec_power_mg][quec_print_os_pm_state, 791] source CP not allow sleep
13:59:21.831	65035	QUEC/I	[quec_power_mg][quec_print_os_pm_state, 791] source USB2 not allow sleep
13:59:24.799	48650	QUEC/I	[quec_power_mg][quec_print_os_pm_state, 791] source CP not allow sleep
13:59:24.819	48651	QUEC/I	[quec_power_mg][quec_print_os_pm_state, 791] source USB2 not allow sleep
13:59:27.798	32267	QUEC/I	[quec_power_mg][quec_print_os_pm_state, 791] source CP not allow sleep
13:59:27.860	32267	QUEC/I	[quec_power_mg][quec_print_os_pm_state, 791] source USB2 not allow sleep
13:59:30.813	15883	QUEC/I	[quec_power_mg][quec_print_os_pm_state, 791] source CP not allow sleep
13:59:30.813	15883	QUEC/I	[quec_power_mg][quec_print_os_pm_state, 791] source USB2 not allow sleep

3.2.2. Matters Needing Attention for Sleep Mode

In sleep mode, if the flow control of main UART is not enabled, data more than 127 bytes cannot be sent directly to the module through the main UART, and can only be sent after exiting sleep mode, otherwise it may be lost.

4 Appendix References

Table 3: Related Document

Document Name
[1] Quectel_EC200U&EG91xU_Series_QuecOpen(SDK)_Quick_Start_Guide

Table 4: Terms and Abbreviations

Abbreviation	Description
API	Application Programming Interface
CSDK	Customized Software Development Kit
IoT	Internet of Things
RRC	Radio Resource Control
RTOS	Real-Time Operating System
USB	Universal Serial Bus